Chapter 2: Arithmetic Expressions

1. **Identify the odd one out from the following expressions, after INTERCHANGING their operators.**

   a) 204 + 3 ÷ 29   b) 67 - 2 × 37   c) 48 ÷ 6 × 3   d) 195 × 49 - 2

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2. **There is a seminar from 11:00 AM to 2:00 PM, where students must spend AT LEAST 1 hour in the seminar. The table displays the number of students entering and exiting the seminar at different times. What is the MAXIMUM possible number of students who spent at least 2 hours in the seminar?**

SR. NO	TIME	IN	OUT
1.	11:00 AM	120	
2.	12:00 PM	40	60
3.	1:00 PM	80	40
4.	2:00 PM		140

   a) 60   b) 80   c) 100   d) 120

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3. **In the given expression, '8 @ 7 - 15 < 4 × 2 × 5' which operator cannot be used in place of '@', such that the condition remains valid?**

   a) -   b) ÷   c) +   d) ×

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4. **A bus has 40 seats, and some seats are already occupied, with one passenger in each seat.**
   - **At the first stop, 5 passengers get off and 8 passengers get in**
   - **At the second stop, 2 passengers get off and 4 passengers get in**
   - **At the third stop, exactly half of the passengers get off**

   After this, half of the bus seats are occupied.
   Which of the following expressions represents the number of passengers on the bus at the beginning?

   a) 5 × 5 + 5   b) 6 × 6 - 4   c) 5 × 5 + 10   d) 6 × 6 - 5

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5. **The following expression has a whole number ‘n’. If a blank can either take ‘×4’ or ‘–12’, what is the least possible value of ‘n’?**
   Expression: (n ____ ____) ____ = 96

   a) 9   b) 18   c) 4   d) 72

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6. **Which operators should be interchanged to get the minimum value of the given expression?**
   8 × 4 + 2 - 1

   a) × and +   b) + and -   c) - and ×   d) No change

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7. **In certain code, 1 is coded as 2, 3 is coded as 10, 5 is coded as 26.**
   In the same way, 4 is coded as A and 6 is coded as B.
   If C is the sum of A and B, which of the following is correct regarding C?

   a) C > 55   b) C < 50   c) C = 50   d) C < 55

8. In the following expression, each ball with a mathematical operator is swapped with the ball immediately to its left, to form a new arithmetic expression. What will be the value of NEW EXPRESSION - ORIGINAL EXPRESSION?

a) 63 b) 73 c) - 63 d) - 73

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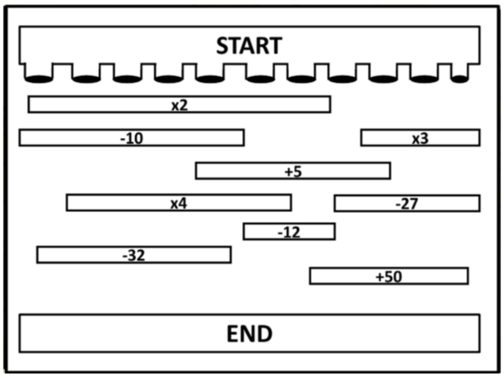
9. A stone with an initial value of 10 is dropped through one of the holes at the START and moves in a straight path to reach the END. As it passes each numbered bar along the way, its value changes according to the operation shown on that bar. What is the maximum value the stone can have when it reaches the END?



a) 60 b) 88 c) 53 d) 68

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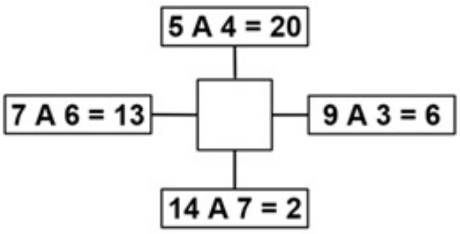
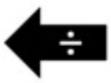
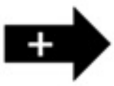
10. Each arrow has a mathematical operator. When placed in the centre box, its operator replaces the letter A in the equation it points to. Which arrow satisfies an equation based on this rule?



a) 

b) 

c) 

d) 

The Thinking Spot

Tom, Sam, Mariya, John, and Bob each selected a shape. They placed their shapes in the grid below.

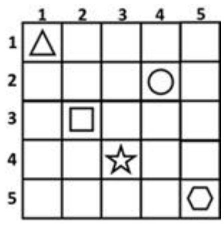
Tom's shape is neither a circle nor a triangle

Sam's shape is in column number 3

Mariya's shape is not in the first three rows

John's shape is neither square nor circle

Which of the following shapes is chosen by Bob?



(a) Square (b) Circle (c) Hexagon (d) Triangle